

# Rahul Rakesh

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## EDUCATION

### University of Western Ontario

London, Ontario

*B.Sc. Honours Specialization in Computer Science (CO-OP)*

2023 – 2028

- Relevant Coursework: Data Structures & Algorithms, Databases, Software Engineering, Operating Systems

## EXPERIENCE

### Freelance Web Developer

Sept. 2024 – Present

*Self-Employed*

*Remote*

- Delivered production-grade full-stack applications for 5+ clients using React, Node.js, Express.js, and PostgreSQL with RESTful API design
- Improved website performance with 50% faster load times and 40% higher engagement through code optimization, lazy loading, and SEO best practices
- Led projects end-to-end: requirements gathering, system design, CI/CD pipelines, and deployment to AWS and Vercel

### Ultimate Frisbee Captain / Coach

June 2022 – Aug. 2025

*School / Merciful Redeemer Parish*

*Mississauga, Ontario*

- Led and coached a competitive team of 15+ players to 2 tournament championships, coordinating training, strategy, and logistics
- Achieved 90% year-over-year player retention through structured training programs emphasizing skill development and teamwork

## PROJECTS

### Fake Out | *React, TypeScript, Node.js, Socket.IO, Docker, Fly.io*

- Built a real-time multiplayer game with sub-100ms latency and 99.5% uptime using React, TypeScript, Node.js, and Socket.IO
- Designed authoritative server-side game state with room-based events and heartbeat monitoring, supporting 10 concurrent players without desynchronization
- Deployed containerized services via Docker on Fly.io with CI/CD pipelines, handling 1,000+ game sessions
- Implemented graceful reconnection logic and session persistence so players rejoin seamlessly after network drops

### Schedula | *Azure Functions, Azure SQL, JWT, REST API*

- Built a serverless booking API with sub-200ms response times using Azure Functions, Azure SQL, and JWT authentication
- Prevented race conditions and double-bookings through transaction-safe scheduling, SQL isolation levels, and idempotent design
- Designed a clean RESTful interface with structured error handling, input validation, and API versioning for maintainability at scale
- Modeled a normalized relational schema with indexed foreign keys to support efficient multi-user concurrent queries under load

### VitaLink | *React Native, Django REST, PostgreSQL | TerraHacks 2025*

- Reduced patient intake time by 30% with a full-stack healthcare platform built in React Native and Django REST Framework
- Designed a normalized PostgreSQL schema with relational data models to support complex patient and scheduling workflows
- Integrated third-party APIs into backend service pipelines, processing structured outputs for downstream features and form pre-filling
- Implemented role-based access control with token authentication to enforce data privacy across patient, provider, and admin tiers

## TECHNICAL SKILLS

**Languages:** Python, Java, JavaScript, TypeScript, C/C++, SQL

**Web & Backend:** React, Node.js, Express.js, Django REST Framework, REST APIs, Socket.IO, React Native

**Databases & Caching:** PostgreSQL, MongoDB, SQLite, Azure SQL (schema design, indexing, query optimization)

**Cloud & DevOps:** AWS (EC2, S3, Lambda), Azure Functions, Docker, CI/CD (GitHub Actions), Vercel, Fly.io